Name:

Collaborators:

Instructions: Worksheets are graded mostly on completion and partially on correctness. Please write complete solutions showing explanations and key steps to the following problems, unless it says otherwise.

Using Lagrange's Method

1. The Method of Variation of Parameters

The goal of the *Variation of Parameters* is to find the particular solution of a *linear 2nd-order ODE with constant coefficients* in the form

$$y'' + py' + qy = f(t)$$

where p and q are constants and $f(t)$ is some function.

Consider this *linear 2nd-order ODE with constant coefficients*

$$y'' - 2y' + y = f(t)$$

where $f(t)$ is some function.

Let $f(t) = e^t$. The following tasks outlines the method of Variation of Parameters to find the particular solution of the 2nd-order linear non-homogeneous ODE.

Task	Show work here
Solve for the homogeneous solution y_h. Then, identify y_1 and y_2.	
—	———
Compute $W = y_1 y_2' - y_2 y_1'$.	
—	———
Evaluate $u_1 = \int \frac{f(t)}{W} y_2 \, dt$.	
—	———
Evaluate $u_2 = \int \frac{f(t)}{W} y_1 \, dt$.	
—	———
Write the particular solution $y_p = -y_1 u_1 + y_2 u_2$. Then, simplify.	

The W term is called the *Wronskian*.

2. An Unusual Non-Homogeneous Term

Consider the same ODE from Problem (1), $y'' - 2y' + y = f(t)$ where $f(t)$ is some function.

Let $f(t) = \frac{e^t}{t}$. In this case, $f(t)$ is not in the list of common trial solutions for Undetermined Coefficients. The Variation of Parameters method must be used to find the particular solution.

a. Using the homogeneous solution in Problem (1a), identify the independent solutions y_1 and y_2. Then, compute the Wronskian

$$W = y_1 y_2' - y_2 y_1'.$$

b. Evaluate each integral using the Wronskian from Part (a):

i. $u_1 = \int \frac{f(t)}{W} y_2 \, dt$

ii. $u_2 = \int \frac{f(t)}{W} y_1 \, dt$

c. Write the particular solution

$$y_p = -y_1 u_1 + y_2 u_2.$$

Then, simplify.

d. Write the general solution $y = y_h + y_p$. Then, solve for the specific solution using the conditions $y(1) = 1$ and $y'(1) = 0$.